



PROGRAMMING IN JAVA

Assignment -- 0

TYPE OF QUESTION: MCQ

Number of questions: 15

Total mark: $15 \times 1 = 15$

QUESTION 1:

What is the data type that occupies the least storage in C language?

- a. float
- b. char
- c. double
- d. int

Correct Answer: b

Detailed Solution:

Character data type allows a variable to store only one character which is of one byte, that is, eight bits. In general, int takes 2 bytes, float takes four bytes and double takes eight bytes.

QUESTION 2:

What is the output of the code segment in C?

```
#include <stdio.h>
int main()
{
    enum Nptel { July = 0, Aug, Dec};
    enum Nptel course = Dec;
    if( course == 0)
        printf("course is in July");
    else if(course == 1)
        printf("course is in Aug");
    if(course == 2)
        printf("course is in Dec");
}
```

- a. course is in July
- b. course is in Aug
- c. course is in Dec
- d. Compilation error



Correct Answer: c

Detailed Solution:

Enumeration data type consists of named integer constants as a list. It start with 0 (zero) by default and value is incremented by 1 for the sequential identifiers in the list.

QUESTION 3:

What is the output of this program?

```
#include <stdio.h>

int main() {
    int i;

    for (i=0;i<10;i++);
    {
        printf("%d ",i);
    }
}
```

- a. 0 1 2 3 4 5 6 7 8 9
- b. 10
- c. 9
- d. Runtime Error

Correct Answer: b

Detailed Solution:

Due to semicolon, for loop is in self loop and then i value is printed.

QUESTION 4:

What does the following declaration mean?

```
int (*ptr)[10];
```

- a. ptr is array of pointers to 10 integers.
- b. ptr is a pointer to an array of 10 integers.
- c. ptr is an array of 10 integers.
- d. ptr is a pointer to a two-dimensional array.



Correct Answer: b

Detailed Solution:

This is the declaration of pointer to an array of size 10.

QUESTION 5:

In C, if you pass an array as an argument to a function, what actually gets passed?

- a. Value of elements in array.
- b. First element of the array.
- c. Base address of the array.
- d. Address of the last element of the array.

Correct Answer: c

Detailed Solution:

When we pass an array as a function argument, the base address of the array will be passed.

QUESTION 6:

Which of the following statements(s) is/are **not true**?

- a. Assembler translates a program in high level programming language into a code in an assembly language.
- b. Compiler translates a program in high level programming language into a code in a machine language.
- c. Interpreter translates a program in high level programming language into a code in a machine language.
- d. A program in C is compiled whereas a program in C++ is interpreted.

Correct Answer: a, d

Detailed Solution:

All high level language (in general) is compiled to code in machine level language. The same strategy is followed in both C and C++ programming languages.

QUESTION 7:

Choose the most appropriate statement(s) in the following.



- a. C, C++ and Java all are high-level programming languages.
- b. C and C++ both support pointer to reference a location of any variable.
- c. A C program is a collection of functions whereas C++ is a collection of both functions and classes.
- d. C provides dynamic memory allocation whereas C++ does not support the same.

Correct Answer: a, b

Detailed Solution:

C program is a collection of functions whereas C++ is a collection of classes. C provides malloc() and calloc() functions for dynamic memory allocation, and free() for memory de-allocation. C++ provides new operator for memory allocation and delete operator for memory de-allocation.

QUESTION 8:

Before the execution of a program, an operating system should

- a. load the program into cache memory.
- b. load the program into register in CPU.
- c. load the program into primary memory.
- d. Run the program from secondary memory

Select the correct option(s) in the above.

Correct Answer: c

Detailed Solution:

The computer can manipulate only data that is in main memory. So, every program you execute and every file you access must be copied from a storage device into main memory.

QUESTION 9:

Consider the following piece of code in C programming language.

```
#include <stdio.h>
int main() {
    int i=10;
    for(int i=0;i<=20;i++)
        i++;
    printf("%d",i);
    return 0;
}
```



If you run this program, then which of the following statement(s) is (are) true?

- a. The program will print the value of i as 10.
- b. The program will print the value of i as 20.
- c. The program will print the value of i as 21.
- d. The loop will execute exactly 10 times.

Correct Answer: a

Detailed Solution:

The i in for loop has local scope and it is within the for block. The variable i outside the block is the scope of the variable in the printf statement.

QUESTION 10:

What is the output of the following code segment?

```
int i = 1;
printf("%d + %d", i++, ++i);
```

- a. 1 + 2
- b. 2 + 3
- c. 2 + 2
- d. 1 + 1

Correct Answer: b

Detailed Solution:

It is the concept of pre-increment and post-increment. The evaluation is done from right-to-left and printf will print the value from left-to-right.

QUESTION 11:

To store the string "Java", how many bytes that a C compiler will take?

- a. 4 bytes.
- b. 5 bytes.
- c. 8 bytes.
- d. 9 bytes.

Correct Answer: b



Detailed Solution:

A C string is as long as the number of characters between the beginning of the string and the terminating null character. So it will take 5 bytes.

QUESTION 12:

Which of the following statement(s) to declare an array, say 'a' is(are) **not** valid in C programming language?

- a. `int a[m];` //m is an integer variable
- b. `int *a;`
- c. `int a[ ]={1,2,3,4,5};`
- d. `int a[10000];`

Correct Answer: a, b

Detailed Solution:

C language needs the size of the array at the time of compilation. Since the value of m is not known at the time of compilation, C compiler will give compilation error. Further, note that *a is a pointer to an integer variable, it can point to an array but not actually an array.

QUESTION 13:

Which of the following is not a valid keyword in C?

- a. auto
- b. void
- c. switch
- d. malloc

Correct Answer: d

Detailed Solution:

1. Keywords are those words whose meaning is already defined by Compiler
2. Cannot be used as **Variable Name**
3. There are **32 Keywords** in C
4. C Keywords are also called as **Reserved words**

The name "malloc" stands for memory allocation. It is not a keyword, rather is a library function defined in <stdlib.h> in C programming language. The malloc() function reserves a block of



memory of the specified number of bytes. And, it returns a pointer of type void which can be casted into pointer of any form.

QUESTION 14:

What result the following macro will produce in a statement in C programming language?

```
#include <stdio.h>
#define square(x)  x*x
int main() {
    int x=36/square(6);
    printf("%d",x);
    return 0;
}
```

- a. 36
- b. 1
- c. 0
- d. 6

Correct Answer: a

Detailed Solution:

The calculation is $36/6*6$. This will be equivalently $(36/6)*6$ because both / and * are of same precedence and with left-to-right associativity.

QUESTION 15:

Which of the following language is not based on object-oriented paradigms?

- a. C
- b. C++
- c. Java
- d. Python

Correct Answer: a

Detailed Solution:

C is not based on 'object-oriented' paradigm rather than function-oriented paradigm. It has no concept of classes, objects, polymorphism, and inheritance.
